THE SHEAF OF DIFFERENTIALS OF R OVER U VIA T AS A QUOTIENT OF THE PULLBACK OF THE CONORMAL SHEAF OF THE IDENTITY SECTION, WITH FLATNESS UPGRADING TO AN ISOMORPHISM

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Allowed standard prerequisites (editorial).

- A section of a morphism of schemes is an immersion (schemes chapter, Lemma schemes-lemma-section-immersion, canonical tag 01KT). cited at proof lines 344-347. Makes e : U to R an immersion, so its conormal sheaf C_{U/R} exists.
- The pullback diagram of the groupoid recording c as a base change of s (more-groupoids chapter, equation equation-pull, canonical tag 04LG; kind equation). cited at proof lines 366-368. Its properties show c is a base change of s, transferring flatness of s to c.
- Functoriality of the conormal sheaf with flat base changes, giving the surjection on conormal sheaves (morphisms chapter, Lemma morphisms-lemma-conormal-functorial-flat, canonical tag 0473). cited at proof lines 363-366. Produces the canonical surjection t^*C_{U/R} to Omega_{R/U} and its upgrade to an isomorphism when c is flat.
- Flatness is preserved under base change (morphisms chapter, Lemma morphisms-lemma-base-change-flat, canonical tag 01U9). cited at proof lines 368-369. From c being a base change of s, flatness of s gives flatness of c, completing the flat case.

Source reference. Original lemma 01KT: sources/stacks-chow/tags.txt, source lines 2062–2062. Cited at proof lines 344-347: e : U to R is an immersion being a section of s.

Source reference. Original equation 04LG: sources/stacks-chow/tags.txt, source lines 6007–6007. Cited at proof lines 366-368: the properties of the pullback diagram show c is a base change of s.

Source reference. Original lemma 0473: sources/stacks-chow/tags.txt, source lines 5494–5494. Cited at proof lines 363-366: produces the canonical surjection t^*C_{U/R} to Omega_{R/U}, an isomorphism when c is flat.

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Source reference. Original lemma 01U9: sources/stacks-chow/tags.txt, source lines 2402–2402. Cited at proof lines 368-369: flatness of s transfers to c along the base change.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. The sheaf of differentials of R seen as a scheme over U via t is a quotient of the pullback via t of the conormal sheaf of the immersion $e : U \to R$. In a formula: there is a canonical surjection $t^*\mathcal{C}_{U/R} \to \Omega_{R/U}$. If s is flat, then this map is an isomorphism.*

Proof. Note that $e : U \to R$ is an immersion as it is a section of the morphism s, see Schemes, Lemma 01KT. Consider the following diagram

$$\begin{array}{ccccc} R & \xrightarrow[(1,i)]{} & R \times_{s,U,t} R & \xrightarrow[(\mathrm{pr}_0, i\circ\mathrm{pr}_1)]{} & R \times_{t,U,t} R \\ {\scriptstyle t}\downarrow & & \downarrow{\scriptstyle c} & & \\ U & \xrightarrow{e} & R & & \end{array}$$

The square on the left is cartesian, because if $a \circ b = e$, then $b = i(a)$. The composition of the horizontal maps is the diagonal morphism of $t : R \to U$. The right top horizontal arrow is an isomorphism. Hence since $\Omega_{R/U}$ is the conormal sheaf of the composition it is isomorphic to the conormal sheaf of $(1, i)$. By Morphisms, Lemma 0473 we get the surjection $t^*\mathcal{C}_{U/R} \to \Omega_{R/U}$ and if c is flat, then this is an isomorphism. Since c is a base change of s by the properties of Diagram (04LG) we conclude that if s is flat, then c is flat, see Morphisms, Lemma 01U9. □